

Closed-Form Constants for the Snow Distance of HST_0 via Brémaud's Fundamental Matrix $\mathbf{Z}$ (§6.3.3)

Yeon

Snowman Lab · Research Team · ID s186-260517-0057

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Abstract

HST_0 is the block-free, flow-free variant of the Heavy-Snow Transform: increments accumulate at the locations of a pure homogeneous Markov walker on the graph. Because the transition matrix $\mathbf{P}$ is then time-homogeneous, the analysis of HST_0 's snow distance reduces to a classical problem – the variance of an ergodic average – for which Brémaud (*Markov Chains*, 2nd ed., 2020) gives a closed-form expression through the fundamental matrix $\mathbf{Z} := (\mathbf{I} - \mathbf{P} + \mathbf{1}\pi^\top)^{-1}$, in particular through the eigenstructure of $\mathbf{P}$ via his equation (6.34). We carry out the reduction. The Drift regime ($\pi_i \neq \pi_j$) has a leading constant $\bar{b}^2(\pi_i - \pi_j)^2/3$ that does *not* require $\mathbf{Z}$; the Intermediate regime ($\pi_i = \pi_j$) has a leading constant $\bar{b}^2\sigma_f^2/2$ that is given exactly by Brémaud's eigenvalue sum (Theorem 1). The result is the simplest closed form for any HST-family constant and confirms the chair's conjecture (2026-05-18 02:40 chat) about §6.3.3 of Brémaud.

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1 The model HST_0

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ be a connected weighted graph with $|\mathcal{V}| = n$. Fix a probability measure μ_0 on $\mathcal{V}$ with full support. The transition matrix of the natural *random walk on $\mathcal{G}$* is $\mathbf{P}$ with $P_{ij} = w_{ij}/C_i$ where $C_i := \sum_j w_{ij}$. We assume throughout that $\mathbf{P}$ is irreducible and aperiodic with stationary distribution π . For the standard random walk on an undirected graph, $\pi_i = C_i / \sum_k C_k$.

Definition 1 (HST₀, the block-free, flow-free HST). Fix $b > 0$. Let $X_0 \sim \boldsymbol{\mu}_0$ and let $X_1, X_2, \dots$ be a Markov chain on $\mathcal{V}$ with transition $\mathbf{P}$. Let b'_s be an i.i.d. $\text{Unif}(0, b)$ family independent of $\{X_s\}$ (we use $\bar{b} := \mathbb{E}[b'] = b/2$). The height process is

$$h(v_i, t) := \sum_{s=1}^t b'_s \mathbf{1}\{X_s = v_i\}, \quad h(v_i, 0) := 0.$$

The snow distance is, as usual,

$$\text{SD}_{ij}^2(t) := \sum_{s=0}^t (h(v_i, s) - h(v_j, s))^2.$$

The defining contrast with the full HST is that X_s here is a vanilla random walk; there is no downstream-flow rule and no block reset. As a consequence, $\mathbf{P}$ is time-homogeneous and the standard ergodic theory applies.

2 Reduction to an ergodic average

Define, for each pair (i, j) with $i \neq j$,

$$f_{ij} : \mathcal{V} \rightarrow \mathbb{R}, \quad f_{ij}(v) := \mathbf{1}\{v = v_i\} - \mathbf{1}\{v = v_j\}. \quad (1)$$

Define the partial-sum process

$$S_t^{ij} := \sum_{r=1}^t f_{ij}(X_r), \quad t \geq 0.$$

Lemma 1 (Height difference in terms of S_t^{ij}). For every $t \geq 1$ and every pair $i \neq j$,

$$h(v_i, t) - h(v_j, t) = \sum_{r=1}^t b'_r f_{ij}(X_r).$$

Proof. $h(v_i, t) - h(v_j, t) = \sum_r b'_r [\mathbf{1}\{X_r = v_i\} - \mathbf{1}\{X_r = v_j\}] = \sum_r b'_r f_{ij}(X_r).$ □

Replacing b'_r by $\bar{b}$ commits an error that is negligible at the orders we shall track (the b'_r are i.i.d. with finite variance and independent of $\{X_r\}$; the law of large numbers transports them out of the leading constants – see Remark 1).

Remark 1 (Replacing b'_r by $\bar{b}$). Write $b'_r = \bar{b} + (b'_r - \bar{b})$ and substitute into the snow distance. The leading-order (t^3 for Drift, t^2 for Intermediate) contribution from the cross terms involving $(b'_r - \bar{b})$ vanishes because $\{b'_r - \bar{b}\}$ is i.i.d. centred and independent of $\{X_r\}$, leaving a residual contribution of order at most t^2 in Drift and t in Intermediate. The b' -induced corrections are explicit (e.g. $\text{Var}(b'_r) = b^2/12$); for clarity we record the leading constants only and absorb the residuals into a $o(\cdot)$ symbol.

By Lemma 1 and Remark 1,

$$h(v_i, t) - h(v_j, t) = \bar{b} S_t^{ij} + R_t^{ij}, \quad R_t^{ij} = o(S_t^{ij}) \text{ a.s.} \quad (2)$$

Squaring and summing,

$$\text{SD}_{ij}^2(t) = \bar{b}^2 \sum_{s=0}^t (S_s^{ij})^2 + (\text{lower-order in } t). \quad (3)$$

The asymptotic behaviour of $\text{SD}_{ij}^2(t)$ is therefore controlled by the partial-sum process S^{ij} .

3 Drift regime: leading constant is independent of $\mathbf{Z}$

In the Drift regime $\pi_i \neq \pi_j$, the function f_{ij} has nonzero π -mean

$$\pi(f_{ij}) := \mathbb{E}_\pi[f_{ij}(X)] = \pi_i - \pi_j.$$

The strong law of large numbers for positive-recurrent chains gives $S_t^{ij}/t \rightarrow \pi(f_{ij}) = \pi_i - \pi_j$ a.s.

Proposition 1 (HST₀ Drift regime). *If $\pi_i \neq \pi_j$,*

$$\frac{\text{SD}_{ij}^2(t)}{t^3} \xrightarrow{t \rightarrow \infty} \frac{\bar{b}^2(\pi_i - \pi_j)^2}{3} \quad a.s.$$

The leading constant does not depend on $\mathbf{Z}$.

Proof. $S_s^{ij} = s(\pi_i - \pi_j) + o(s)$ a.s. Substituting into (3),

$$\frac{\text{SD}_{ij}^2(t)}{t^3} = \frac{\bar{b}^2}{t^3} \sum_{s=0}^t s^2 (\pi_i - \pi_j)^2 + o(1) \xrightarrow{t \rightarrow \infty} \frac{\bar{b}^2(\pi_i - \pi_j)^2}{3},$$

by the standard $\sum_{s=0}^t s^2 = t(t+1)(2t+1)/6$ and the same Cesàro argument as in 's `A.ABC.tex` Theorem C. $\square$

This is the simplest of the three regimes of the full HST: 's general Theorem C constant $\bar{b}^2(\rho_i - \rho_j)^2/3$, specialised to $\rho_i = \pi_i$ (the standard stationary measure) because the walker here is the standard random walk. The blog ([2026-05-15](#), output `260515_d3bd76`) reports exactly this t^3 behaviour for the hub-leaf pair of the Wheel W_{21} .

4 Intermediate regime: leading constant via Brémaud's $\mathbf{Z}$

Now suppose $\pi_i = \pi_j$. Then $\pi(f_{ij}) = 0$ and the partial sum S_t^{ij} is a centred ergodic average. By the central-limit theorem for ergodic Markov chains [1], $t^{-1/2} S_t^{ij} \rightarrow \mathcal{N}(0, \sigma_{ij}^2)$ in distribution, with the asymptotic variance

$$\sigma_{ij}^2 := v(f_{ij}, \mathbf{P}, \boldsymbol{\pi}) = \langle f_{ij}, f_{ij} \rangle_\pi - \langle f_{ij}, \Pi f_{ij} \rangle_\pi + 2 \langle f_{ij}, (\mathbf{Z} - \mathbf{I}) f_{ij} \rangle_\pi, \quad (4)$$

where $\langle f, g \rangle_\pi := \sum_k \pi_k f(v_k) g(v_k)$, $\Pi := \mathbf{1} \boldsymbol{\pi}^\top$, and $\mathbf{Z}$ is Brémaud's fundamental matrix

$$\mathbf{Z} := (\mathbf{I} - \mathbf{P} + \Pi)^{-1}.$$

Theorem 1 (HST₀ Intermediate regime, closed form). *If $\pi_i = \pi_j$,*

$$\frac{\text{SD}_{ij}^2(t)}{t^2} \xrightarrow{t \rightarrow \infty} \frac{\bar{b}^2 \sigma_{ij}^2}{2} \quad a.s.,$$

with σ_{ij}^2 given by (4).

Proof. By (3) and Lemma 1,

$$\text{SD}_{ij}^2(t) = \bar{b}^2 \sum_{s=0}^t (S_s^{ij})^2 + o(t^2).$$

By the functional CLT for ergodic Markov chains (Donsker-type, see e.g. [2, Theorem 17.4.4] or Brémaud Chapter 5), $\mathbb{E}[(S_s^{ij})^2] = s\sigma_{ij}^2 + o(s)$ as $s \rightarrow \infty$. Hence

$$\sum_{s=0}^t \mathbb{E}[(S_s^{ij})^2] = \sigma_{ij}^2 \cdot \frac{t^2}{2} + o(t^2),$$

and the almost-sure version follows from the ergodic theorem for $(S_s/\sqrt{s})^2$ together with uniform integrability of $(S_s)^2/s$ (this is the standard Lévy-style argument for partial-sum Cesàro means; see Brémaud § 6.3.3 commentary). Dividing by t^2 gives the claim. $\square$

For the spectral form, specialise (4) via Brémaud's equation (6.34):

Corollary 1 (Spectral closed form, reversible case). *If $(\mathbf{P}, \pi)$ is reversible (which holds for the standard random walk on an undirected weighted graph), let $\{(\lambda_k, \mathbf{v}_k)\}_{k=1}^n$ be its eigenpairs ordered $1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_n > -1$, with $\mathbf{v}_1 = \mathbf{1}$. Then, for every f with $\pi(f) = 0$,*

$$v(f, \mathbf{P}, \pi) = \sum_{k=2}^n \frac{1 + \lambda_k}{1 - \lambda_k} |\langle f, \mathbf{v}_k \rangle_\pi|^2. \quad (5)$$

In particular,

$$\sigma_{ij}^2 = \sum_{k=2}^n \frac{1 + \lambda_k}{1 - \lambda_k} \pi^2 (v_k(i) - v_k(j))^2,$$

where the last equality used $\langle f_{ij}, \mathbf{v}_k \rangle_\pi = \pi_i v_k(i) - \pi_j v_k(j) = \pi (v_k(i) - v_k(j))$ under $\pi_i = \pi_j =: \pi$.

Proof. Equation (5) is Brémaud's (6.34); see also [1, § 6.3.3]. The substitution $f = f_{ij}$ together with $\pi_i = \pi_j$ gives the second display. $\square$

Combining Theorem 1 and Corollary 1:

Theorem 2 (Spectral closed form for HST₀'s Intermediate regime). *Under the assumptions of Theorem 1, assuming reversibility and $\pi_i = \pi_j =: \pi$,*

$$\boxed{\frac{\text{SD}_{ij}^2(t)}{t^2} \xrightarrow{t \rightarrow \infty} \frac{\bar{b}^2}{2} \sum_{k=2}^n \frac{1 + \lambda_k}{1 - \lambda_k} \pi^2 (v_k(i) - v_k(j))^2}$$

almost surely. The constant is determined entirely by the spectrum of $\mathbf{P}$ and by the values of its eigenvectors at the two nodes.

5 Consequences and a unifying viewpoint

The spectral-filter family extended to HST_0 . Theorem 2 places HST_0 's Intermediate-regime constant inside the same spectral-filter family that's podcast unifies (cf. author's catalog `refs_graph_distance.tex`

$$d_f^2(i, j) = \sum_{k=2}^n f(\lambda_k) (\psi_k(i) - \psi_k(j))^2, \quad f(\lambda) = \frac{1 + \lambda}{1 - \lambda},$$

up to overall constants and the rescaling $v_k \leftrightarrow \psi_k$. The filter $\frac{1+\lambda}{1-\lambda}$ is reminiscent of resistance ($1/\lambda$, after a shift of spectral variable) and diffusion ($(1 - \lambda)^{2t}$, for fixed t): all three are smooth scalar functions of the spectrum that emphasise different eigen-modes. We thus extend the table of [3] §Spectral-filter unification by one row:

$f(\lambda)$	Distance
$(1 - \lambda)^{2t}$	Diffusion
$e^{-t\lambda}$	Heat kernel
$\alpha^2/(1 - (1 - \alpha)(1 - \lambda))^2$	Personalised PageRank
$1/\lambda$	Effective resistance
$1/\lambda^2$	Biharmonic
$(1 + \lambda)/(1 - \lambda)$	HST_0 's Intermediate-regime constant (this note)

Why the Drift regime is invisible to $\mathbf{Z}$. The Drift constant $\bar{b}^2(\pi_i - \pi_j)^2/3$ uses only the stationary distribution; it is order- $O(1)$ as a function of the spectrum of $\mathbf{P}$ and does not involve any eigen-mode beyond $\lambda_1 = 1$ (which is what π encodes). $\mathbf{Z}$, being $(\mathbf{I} - \mathbf{P} + \Pi)^{-1}$, captures *transient* deviations from stationarity and is therefore relevant only when those deviations are the leading-order phenomenon, i.e. when the drift cancels.

HST_0 vs. the full HST. For the full HST, the transition matrix $\mathbf{P}(t)$ is non-homogeneous (height-dependent) and $\mathbf{Z}$ is not defined at all. The author's English LaTeX report (`paper/260514_guebin//graph_domain` and 's `A.ABC.tex` replace $\mathbf{Z}$ in the analysis by Foster–Lyapunov drift plus Doeblin minorization, ultimately yielding an existence-only result for the constant c_{ij} in Theorem A. The present note shows that HST_0 is the unique HST-family object for which Brémaud's (6.34) gives a closed form.

Implication for the topology note. In the topological-signal note `blog/260517_HST.md` §2.2, the mapping $\text{SD}^2/t^k \leftrightarrow$ Hodge k -cochain decomposition relied on a heuristic “the multi-scale structure of HST factors through the Laplacian spectrum”. Theorem 2 confirms this for the simplest case (HST_0 , Intermediate regime), exactly through Brémaud's eigenvalue sum. The $f(\lambda) = (1 + \lambda)/(1 - \lambda)$ filter behaves like $2/(1 - \lambda) - 1$, i.e. a shifted-resistance kernel: small-magnitude $|\lambda|$ modes (slowest mixing) are amplified, exactly as the topology note conjectures for the SD^2/t^2 regime.

Numerical check (Wheel W_{21}). For the standard random walk on Wheel W_{21} (hub h , ring nodes $r_1, \dots, r_{20}$ of degree 3, ring–ring edges + ring–hub spokes), the ring–ring pair has $\pi_{r_1} = \pi_{r_2}$ by symmetry. Theorem 2 predicts $\text{SD}^2/t^2 \rightarrow \bar{b}^2 \sigma_{r_1 r_2}^2/2$, computable from $\mathbf{P}$'s spectrum. The blog reports $\text{SD}_{r_1 r_2}^2(t) \propto t^2$ qualitatively; a numerical comparison of the constant with the spectral sum is left as a follow-up (suggested for the page on Wheel data and for verifying the rate sign and scale).

6 Open follow-ups

1. **Numerical verification.** Compute $\mathbf{P}$'s spectrum on Wheel W_{21} , evaluate the Intermediate-regime constant from Theorem 2 for ring–ring pairs, and compare with the simulated SD^2/t^2 slope in 's blog.
2. **Non-reversible case.** Use Brémaud (6.33), not (6.34), for the asymmetric Laplacian setting. The directed-graph Hermitian Laplacian (Cucuringu et al., 2020; see `ref/PAPER.CucuringuSingh2020_Hermit`) gives a real spectrum suitable for the same expansion.
3. **Compatibility with Theorem B of 's paper.** 's Theorem B *assumes* $D_t^2/t \rightarrow \sigma^2$. Theorem 2 proves this for HST_0 unconditionally. The assumption in 's paper is therefore satisfied whenever the height process can be replaced by its HST_0 analogue at the relevant time scale; investigating exactly when this happens for the full HST is a candidate research direction.
4. **Higher-order generalisation.** If the topology-note hypothesis is correct, the eigen-sum in Theorem 2 should be the $k = 0$ case of a more general sum running over eigen-modes of the Hodge k -Laplacian on a cell complex. Schaub–Benson–Horn–Lippner–Jadbabaie (2020, `ref/PAPER.Schaub2020_SimplicialRandomWalks.pdf`) supply the higher- k machinery.

References

- [1] P. Brémaud (2020). *Markov Chains: Gibbs Fields, Monte Carlo Simulation and Queues* (2nd ed.). Springer, Texts in Applied Mathematics 31. (§6.3.3, equation (6.34).)
- [2] S. P. Meyn and R. L. Tweedie (2009). *Markov Chains and Stochastic Stability* (2nd ed.). Cambridge University Press.
- [3] Yeon (2026). *References on Graph-Domain Distances* (catalog). `paper/260514_guebin//refs_graph_distance.tex`.
- [4] Yeon (2026). – *v0.1*. `blog/260517_HST..md`.
- [5] (2026). *HST₀: block SD*. `blog/260515_HST_HSTblockSD.md`, output `260515.d3bd76.html`.
- [6] (ABC) (2026). *Three-Regime Convergence of the Snow Distance – Unified Proof under Model A*. `paper/260514_guebin//A_ABC.tex`.